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Combinatorial CRISPR screen reveals *FYN* and *KDM4* as targets for synergistic drug combination for treating triple negative breast cancer

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Abstract

Tyrosine kinases play a crucial role in cell proliferation and survival and are extensively investigated as targets for cancer treatment. However, the efficacy of most tyrosine kinase inhibitors (TKIs) in cancer therapy is limited due to resistance. In this study, we identify a synergistic combination therapy involving TKIs for the treatment of triple negative breast cancer. By employing massively parallel combinatorial CRISPR screens, we identify *FYN* and *KDM4* as critical targets whose inhibition enhances the effectiveness of TKIs, such as NVP-ADW742 (IGF-1R inhibitor), gefitinib (EGFR inhibitor), and Imatinib (ABL inhibitor) both *in vitro* and *in vivo*. Mechanistically, treatment with TKIs upregulates the transcription of *KDM4*, which in turn demethylates H3K9me3 at *FYN* enhancer for *FYN* transcription. This compensatory activation of *FYN* and *KDM4* contributes to the resistance against TKIs. We highlight *FYN* as a broadly applicable mediator of therapy resistance and persistence by demonstrating its upregulation in various experimental models of drug-tolerant persisters and residual disease following targeted therapy, chemotherapy, and radiotherapy. Collectively, our study provides novel targets and mechanistic insights that can guide the development of effective combinatorial targeted therapies, thus maximizing the therapeutic benefits of TKIs.

eLife assessment

This study presents a **valuable** finding on the roles and mechanisms of FYN and KDM4 in TNBC tumor cell resistance. The evidence supporting the claims of the authors is somewhat **incomplete** and the refinement of certain experiments would have strengthened the study. Noteworthy, FYN has been implied in drug resistance previously and this should be carefully discussed in the manuscript. The work will be of interest to scientists working on breast cancer.

Introduction

Tyrosine kinases have emerged as important drug targets in cancer therapy due to their druggability and pivotal roles in cell proliferation and survival(1). They are implicated in various aspects of cancer development(2), such as cell survival, proliferation, angiogenesis, and invasion, making them attractive targets for drug intervention. Consequently, tyrosine kinase inhibitors (TKIs) have gained considerable attention as primary agents for cancer treatment.

Triple negative breast cancer (TNBC) treatment has limited options for targeted therapy. TNBC, characterized by the absence of estrogen receptor, progesterone receptor, and HER2 expression, exhibits elevated activity of tyrosine kinases, including EGFR and IGF1R(3, 4). However, several clinical trials investigating TKIs, such as VEGFR inhibitors, EGFR inhibitors, and FGFR inhibitors, in TNBC treatment have yielded disappointing results due to inadequate efficacy. Therefore, it is crucial to comprehend the mechanisms underlying TNBC's suboptimal response to TKIs to enable the development of more effective targeted therapies against TNBC.

The therapeutic efficacy of TKIs is compromised by intrinsic and acquired resistance(5). For instance, EGFR inhibitor gefitinib extended the median progression-free survival by only five months compared to conventional chemotherapy in non-small cell lung cancer (NSCLC) patients with EGFR mutation(6). Significant subset of drug resistance is driven by gene interactions that enable compensatory changes in signal transduction upon drug treatment. Compensatory activation of mitogenic signals, such as MET, PIK3CA amplification, and MAPK/ERK signaling activation, counterbalances the inhibition of EGFR by TKI osimertinib in a significant portion of NSCLC patients(7). Simultaneous inhibition of multiple signaling molecules that compensate for each other's loss is proposed as an effective strategy to overcome resistance to kinase inhibitor therapy, emphasizing the importance of combinatorial therapy(8).

Until recently, a highly scalable method for screening combinatorial therapy has been lacking. Combinatorial CRISPR screens have emerged as efficient tools to identify synergistic targets for combinatorial therapy. We and others recently developed massively parallel combinatorial CRISPR screens to elucidate pairwise gene interactions(9-12). Our combinatorial genetic screen platform, combinatorial genetics *en Masse* (combiGEM), was successfully implemented to identify combinations of epigenetic regulators causing synthetic lethality in ovarian cancer cells (9).

In this study, we utilize CombiGEM-CRISPR technology to identify synergistic tyrosine kinase inhibitor combinations for effective elimination of TNBC. We highlight FYN as a key therapeutic target that, when inhibited, enhances the cytotoxic effect of inhibition of other tyrosine kinases (IGF1R, EGFR, and ABL2). Mechanistic studies reveal KDM4 as a crucial epigenetic regulator that demethylates H3K9me3 and transcriptionally upregulates *FYN* upon TKI treatment. *In vitro* and *in vivo* validation demonstrates the synergistic TNBC-shrinking effects of combining PP2, saracatinib (FYN inhibitor) or QC6352 (KDM4 inhibitor) with TKIs. Additionally, we demonstrate the clinical

significance of our findings by observing upregulation of *FYN* in various models of drug tolerant persisters and residual tumors after chemo-, radio-, or targeted therapy. Therefore, simultaneous targeting of *FYN-KDM4* and tyrosine kinase pathways through combinatorial therapy holds promise for effective therapy against TNBC.

Results

For efficient translation of CRISPR screening data to drug combination, we selected 76 tyrosine kinases that could be inhibited by at least one drug from the drug repurposing hub database (table S1) (13). For pairwise CombiGEM library construction, we chose three guide RNAs from the optimized Brunello sgRNA list (14) employing the iterative cloning method as previously described (9, 15). The resulting library enabled massively parallel screens of pairwise knockouts, encompassing 54,289 sgRNA pairs representing 3,003 pairwise gene disruptions (Fig. 1A). To validate our library, we performed next-generation sequencing and confirmed that 99.5% (2,989/3,003) of gene pairs were represented by at least 6 pairs of sgRNAs, with the log₁₀ reads per million of 0.5 (Fig. S1A).

Triple negative breast cancer cell line MDA-MB-231 cells stably expressing Cas9 were infected with the lentiviral library at low multiplicity of infection (MOI) of 0.3. Genomic DNA was harvested 3 days after infection (designated as day 0[D0]), and 23 days after infection (D20) (Fig. 1A) to perform PCR amplification of sgRNA pairs for subsequent next-generation sequencing (NGS) analysis. Our previous CombiGEM screens, involving the sequencing of contiguous stretches of barcodes for sgRNAs, did not accurately reflect the combinations of sgRNAs expressed in cells due to the uncoupling of sgRNA and barcodes during lentiviral replication and reverse transcription (16). In fact, the decoupling rate of barcode from sgRNA increased with the distance between them, reaching ~35% in pairwise screens (Fig. S1B). To address this issue, we directly sequenced the sgRNA spacer sequences using paired-end sequencing (Fig. S1C). We counted the occurrences of each sgRNA pair in the NGS data and calculated the normalized log₂ fold change in counts between the day 20 and day 0 samples as the growth phenotype score *Z* (see Methods). The counts for the two permutations of an sgRNA pair (e.g., sgRNA-A + sgRNA-B and sgRNA-B + sgRNA-A) exhibited good correlations (*r* = 0.50) and were combined when calculating *Z* (Fig. S1D). The *Z* scores for three biological replicates also demonstrated strong correlations (*r* = 0.74 between replicates #2 and #3) (Fig. S1E). Additionally, we confirmed that independent sgRNA pairs targeting the same set of genes showed correlated changes in the growth phenotype (Fig. S1F-G).

We aimed to identify gene perturbations that exhibit synergistic cell-killing effects by calculating gene interaction scores (GI). The GI scores were derived by comparing the growth phenotype score *Z* resulting from the disruption of a gene pair (*Z*_{A+B}, observed *Z* score) to the sum of *Z* scores obtained from the disruption of each gene individually within the pair (*Z*_{A+Con} + *Z*_{B+Con}, expected *Z* score). We computed GI scores at both the sgRNA level and the gene level. At the gene level, we employed the *Z* scores of all sgRNA pairs targeting the same gene pair to calculate the gene interaction score (GI), as demonstrated in Figure 1B. The expected and observed *Z* scores for each gene (or sgRNA) pair exhibited a strong positive correlation, as shown in Figure 1B for gene level analysis and Figure S1H for sgRNA level analysis. This correlation suggests that most random pairwise combinations of tyrosine kinase perturbations have merely additive effects on cancer cell killing, as expected. To calculate the GI scores, we normalized the observed *Z* score by quantifying its deviation from the quadratic fit of the expected-observed *Z* score plot (10). The GI scores were normalized (Fig. 1C) by dividing them by the standard deviation of the GI scores obtained from the 200 nearest neighbors in terms of expected *Z* scores. The standard deviation of the raw GI scores tend to increase at extremely low expected *Z* scores as previously reported (11) (Fig. S1I).

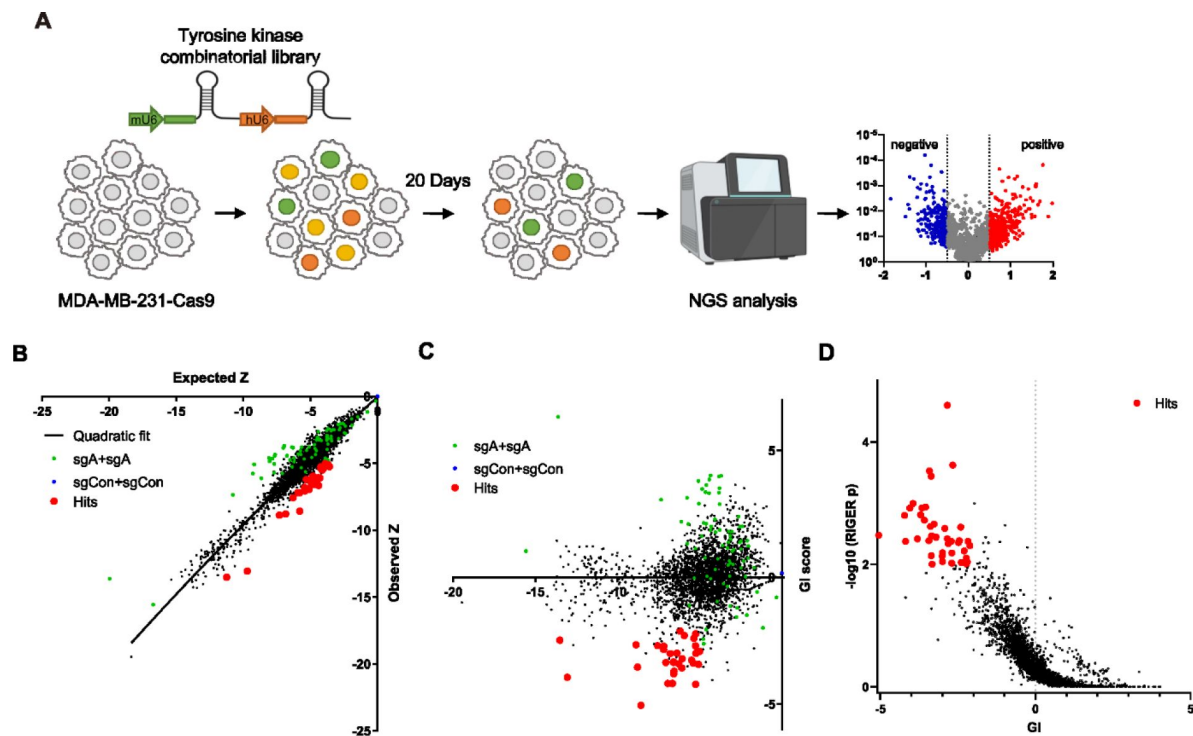


Figure 1.

Pairwise CRISPR screen reveals combinations of synthetic lethal tyrosine kinase ablations.

(A) Schematic diagram of combinatorial screens performed in TNBC cell line MDA-MB-231. (B) Scatter plot of expected growth phenotype Z score and observed growth phenotype Z score of each gene combination. Green dots indicate gene combinations where the identical gene is targeted by the two sgRNA. Red dots indicate candidate synthetic lethal gene pairs listed in table S1. (C) Scatter plot of growth phenotype Z score and normalized GI score of each gene combination. (D) Scatter plot of gene level GI score and RIGER p value calculated with GI scores of each sgRNA pairs that target the given gene pair.

Using cutoffs for gene level GI score <-2 and $p<0.01$ for GI scores determined by RIGER analysis(17) and $Z<-5$ (Fig. 1D and dataset S1), we selected thirty synthetic lethal gene pairs. Among these, the SRC-YES pair exhibited one of the strongest synthetic lethal gene pairings (GI = -3.95). Notably, SRC-YES belong to the same tyrosine kinase family and are known to be functionally redundant and are expected to be synthetic lethal(18). These findings provide evidence for the effectiveness of our screening approach in identifying synthetic lethal gene pairs.

We subsequently conducted validation experiments to confirm the occurrence of synergistic cell death in the candidate synthetic lethal pairs. To achieve this, we introduced lentiviral vectors carrying two distinct single-guide RNAs (sgRNAs) targeting the candidate synthetic lethal pairs, each tagged with a different fluorescent protein (GFP and mCherry). MDA-MB-231-Cas9 cells were infected with the lentivirus at a low titer (MOI ~ 0.5), resulting in a mixed population of cells expressing either one or both sgRNAs along with their respective fluorescent proteins (Fig. 2A). To evaluate synthetic lethality, we monitored the progressive decrease in the number of GFP/mCherry double-positive cells over time (see methods). We validated the efficacy of the sgRNAs used in Figure 2A through the T7 endonuclease assay, which confirmed efficient gene editing (Fig. S2A). Consistent with our CRISPR screening results, we observed that the disruption of six out of eight synergistic target gene combinations led to a reduction in cell viability beyond what was predicted by the Bliss independence model (Fig. 2B). Moreover, the relative viability of double knockout cells and the rate of synergistic killing demonstrated a strong correlation with our screening data ($r = 0.65$ for both viability and synergistic effect; Fig. S2B). Collectively, our findings provide compelling evidence that our screening approach successfully identified synthetic lethal gene pairs with a high level of confidence.

During our analysis, we observed that several validated synergistic target gene pairs included *FYN* (e.g. *FYN*+*IGF1R*, *FYN*+*EGFR*, and *FYN*+*ABL2*). Notably, network analysis of the 30 candidate synergistic tyrosine kinase pairs revealed that *FYN* is one of the key nodes participating in synergistic interactions with multiple genes (Fig. 2C). Interestingly, we found that *FYN*, but not *SRC*, exhibited significant upregulation in triple-negative breast cancer (TNBC) compared to other subtypes, as evidenced by microarray data from primary tumor samples (19) and the cancer cell line encyclopedia (CCLE)(20) (Figs. 2D-E). These findings suggest that gene A could represent an attractive drug target for TNBC treatment. To investigate this further, we assessed whether simultaneous inhibition of *FYN* by PP2, which selectively targets the SRC family kinase inhibitor with the highest potency against *FYN*, in combination with other kinase inhibitors (TKIs), could inhibit cancer cell growth(21). Intriguingly, analysis using SynergyFinder(22) revealed that all TKI combinations involving PP2 and NVP-ADW742 (IGF1R inhibitor), gefitinib (EGFR inhibitor) or Imatinib (ABL inhibitor) synergistically induced cell death in MDA-MB-231 cells (Fig. 2F). Dose-response curves demonstrated that co-treatment with PP2 reduced the IC₅₀ of the tested TKIs by 34-61%, indicating that PP2 sensitized cancer cells to TKI treatment (Fig. 2G). Similar synergy was observed when TKI combinations included saracatinib in place of PP2 (Fig. S3A). Moreover, specific ablation of *FYN*, but not *SRC*, sensitized cells to TKIs, highlighting the critical role of *FYN* as a member of SRC kinase family responsible for TKI resistance (Figs. 2H and S3B). Importantly, we observed similar synergy between the same drug combinations in other TNBC cell lines, including Hs578T, HCC1143, HCC1395, and HCC1937 cells (Fig. S3C-F). Further assessment using live-dead and BrdU assays revealed that both the PP2+NVP-ADW742 and PP2+gefitinib combinations synergistically induced cell death while inhibiting cell growth (Fig. 2I).

Persistent activation of MAPK pathway and PI3K-AKT pathway has been associated with TKI resistance in various cancers(5). Therefore, we investigated which downstream pathways were involved in sensitizing cells to TKI treatment. Notably, the p38 MAPK was significantly attenuated following treatment with either PP2 or saracatinib treatment (Fig. 2J). Genetic ablation of *FYN*

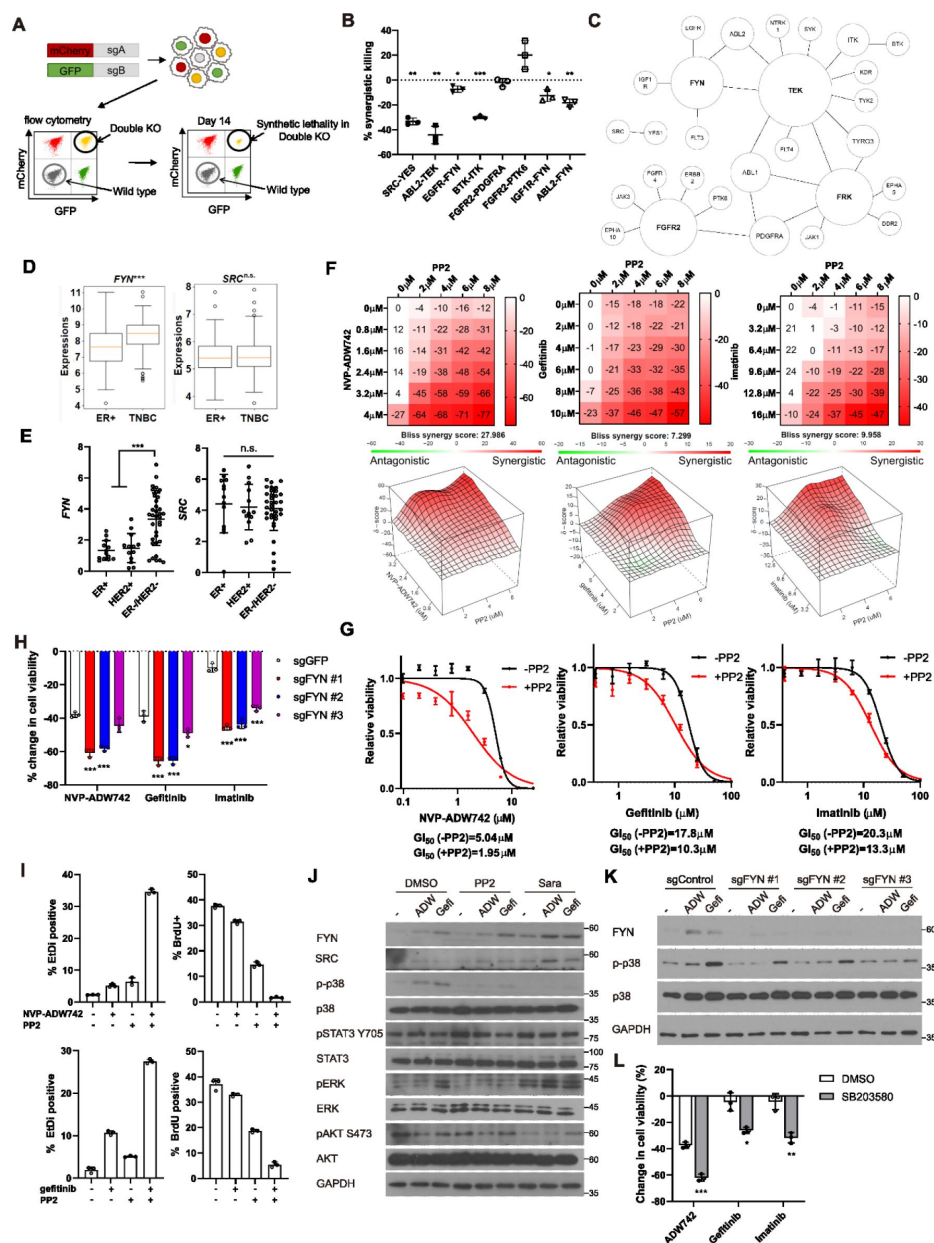


Figure 2.

FYN is critical mediator of TKI resistance.

(A) Schematic diagram of *in vitro* validation of synthetic lethal gene pairs using sgRNAs. (B) Summary of synergistic killing by sgRNAs targeting indicated gene pairs (n=3). (C) Network analysis of the 30 candidate synthetic lethal gene pairs. The size of each node is proportional to the number of connections the gene has. (D-E) *FYN* and *SRC* mRNA expressions in (D) microarray data of primary breast cancers of GSE25066 cohort, and (E) in cancer cell line encyclopedia, for indicated subtypes. (F) Summary of MTT assay with MDA-MB-231 cells treated with the TKI combinations at indicated concentrations (n=2). Synergistic killing is calculated using SynergyFinder with Bliss independence model. (G) Dose response curve of the indicated TKI in the presence and absence of PP2 (n=3). (H) MTT assay with MDA-MB-231 Cas9 cells expressing indicated sgRNAs treated with indicated TKIs (n=3). (I) Cell death and cell proliferation in MDA MB 231 cells treated with NVP-ADW742, gefitinib and PP2 either as single agent or as combination (n=3). (J) Western blot analysis of MDA-MB-231 cells treated with indicated drugs. (K) Western blot analysis of MDA-MB-231 Cas9 cells expressing indicated sgRNA and treated with indicated drugs. (L) MTT assay of MDA-MB-231 cells treated with indicated drugs (n=3). All data are plotted as mean±s.d. One sample t-test for B, and unpaired two-sided Student's t-test in D,E,H and L. *, p<0.05; **, p<0.01; ***, p<0.001; n.s., p>0.05. All replicates are biological replicates.

similarly reduced p38 activation (**Fig. 2K**). Attenuation of p38 activity was also observed in an independent TNBC cell line, Hs578T (Figs. S3G-H). Importantly, treatment of p38 MAPK pathway inhibitor SB203580 markedly sensitized cells to TKI treatment (**Fig. 2L**).

Our discovery that inhibition of FYN synergizes with multiple TKIs possessing distinct target profiles suggests that FYN may play a role in general resistance mechanisms against TKI therapy. Consistently, we observed an increase in both protein and mRNA levels of *FYN* following TKI treatment, indicating that upregulation of FYN confers compensatory survival signal in TKI-treated cells (**Fig. 3A-B**). To elucidate the mechanisms underlying the accumulation of *FYN*, we treated MDA-MB-231 cells with inhibitors targeting key epigenetic modifiers and assessed their synergistic effects with NVP-ADW742 in cell killing, as well as their impact on *FYN* mRNA accumulation. Intriguingly, we found that GSK-J4, an inhibitor of the jumonji domain histone demethylase family (23), was the only drug in our initial screen that decreased *FYN* mRNA and viability upon TKI treatment (Figs. S4A-B). Furthermore, treatment with NVP-ADW742 increased the expression of most members of the jumonji domain histone demethylase family (**Fig. 3C**). This observation is consistent with a previous study on taxane-resistant H1299 lung cancer cells(24), suggesting that histone demethylases may play critical roles in activating a drug resistance gene program. However, the ablation of KDM6, the primary targets of GSK-J4, failed to significantly decrease *FYN* expression (Fig. S4C). GSK-J4 is known to inhibit other jumonji domain histone demethylase family proteins including KDM4 and KDM5(25). Therefore, we tested the possibility that other histone demethylase may be involved in regulating *FYN* expression. Among jumonji domain histone demethylases, *KDM4*, and to a lesser extent *KDM3*, were the only gene family members whose ablation inhibited *FYN* upregulation and p38 activation upon TKI treatment (**Figs. 3D** and S4D). Ablation of KDM5, which has been shown to induce drug tolerance in cancer cells(26), did not significantly alter *FYN* expression (Fig. S4E). Similar to NVP-ADW742 treatment, gefitinib treatment increased *KDM4* demethylase levels (Fig. S4F). We also analyzed two independent TNBC organoids obtained from primary tumors and found concurrent upregulation of *KDM4* with *FYN* mRNAs upon NVP-ADW742 and gefitinib treatment (Fig. S4G). Both KDM3 and KDM4 demethylates methylated H3K9, thereby promoting the opening heterochromatin for transcription(27). Remarkably, *KDM4A* expression was upregulated in TNBC (**Fig. 3E**) and exhibited a positive correlation with *FYN* expression in CCLE database, suggesting that KDM4 regulates *FYN* mRNA levels (**Fig. 3F**). Genetic ablation of *KDM3* or *KDM4* (Fig. S5A) decreased *FYN* and p38 activity, sensitizing MDA-MB-231 cells to TKIs (**Figs. 3G-H**). Likewise, treatment of KDM4 inhibitor QC6352(28) synergized with TKIs in killing MDA-MB-231 cells (**Fig. 3I**). QC6352 treatment also significantly attenuated *FYN* accumulation upon NVP-ADW742 treatment (**Fig. 3J-K**). This was consistent with the RNA sequencing data results in the previous study with breast cancer stem cells treated with QC6352(29). Specifically, *FYN* was the most significantly downregulated SRC family kinase upon QC6352 treatment among (**Fig. 3L**). Analysis of chromatin IP (ChIP) sequencing data from the same study revealed KDM4A enrichment near *FYN* promoter; and QC6352 treatment increased H3K9me3 enrichment at the same locus (**Fig. 3M**). Indeed, this *FYN* promoter locus exhibited a reduction in H3K9me3 following NVP-ADW742 treatment, while QC6352 treatment restored H3K9me3 enrichment (**Fig. 3N**). This finding suggests that KDM4 may directly demethylate H3K9me3 at *FYN* promoter to upregulate *FYN* transcription. *FYN* accumulation and resistance to TKIs were also confirmed to be attenuated by QC6352 treatment in other independent TNBC cell lines (Figs. S5B-C).

We proceeded to investigate the potential clinical application of our synthetic lethal gene pairs as combinatorial therapy by assessing the *in vivo* efficacy of pharmacological interventions targeting these gene pairs using MDA-MB-231 xenograft models. Strikingly, co-treatment of saracatinib and NVP-ADW742 synergistically reduced tumor size, whereas treatment with either agent alone was ineffective in slowing tumor growth (**Fig. 4A**). All treatment groups exhibited minimal changes in body weight, indicating that the overall health of the animals was not adversely affected by the combination treatment (Fig. S6A). Saracatinib-gefitinib combination was not tested as saracatinib can inhibit EGFR(30). Similarly, KDM4 inhibitor QC6352 synergized with gefitinib in reducing

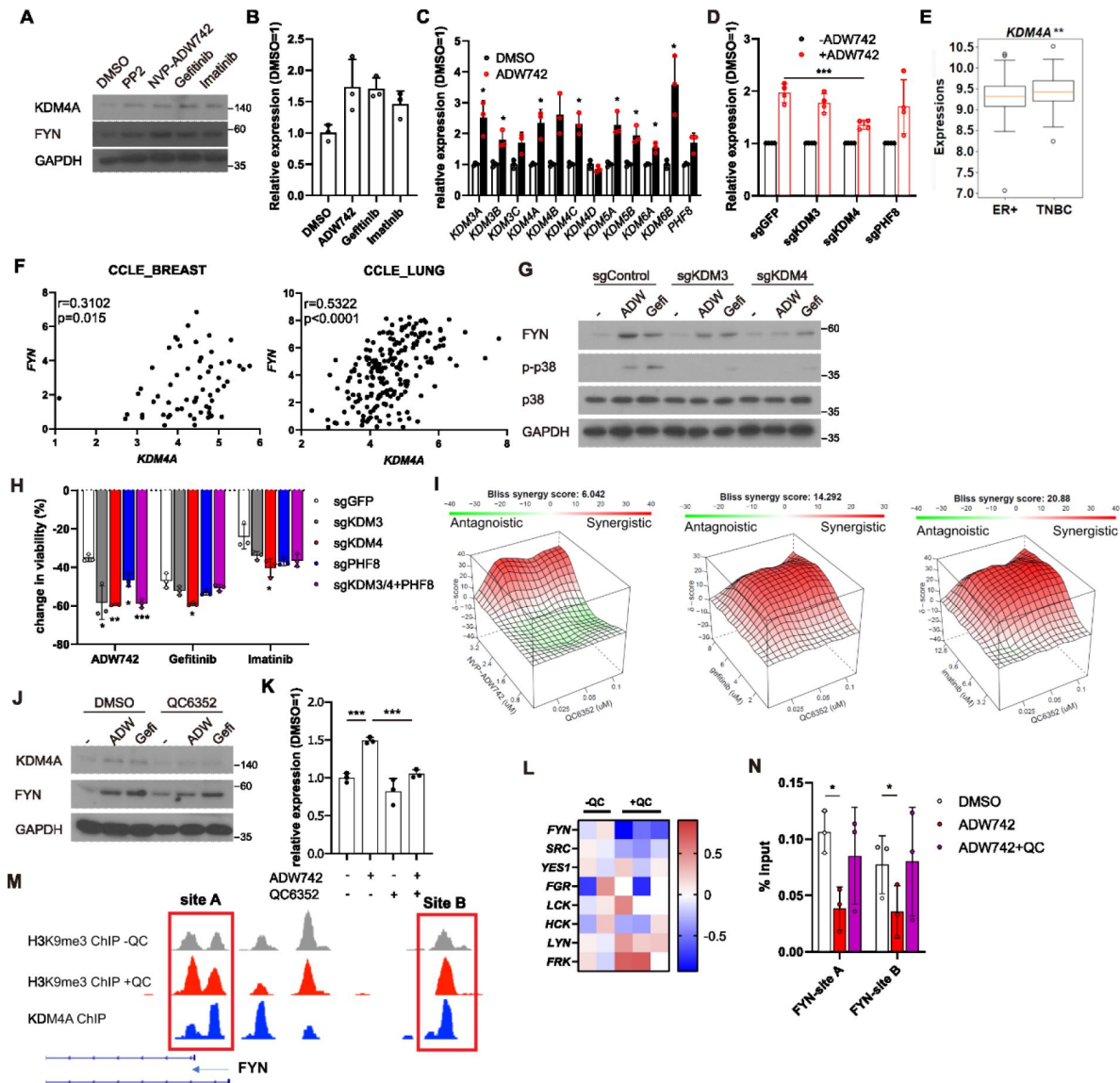


Figure 3.

Activation of *KDM4* upregulates *FYN*, conferring drug resistance.

(A) Western blot analysis of MDA-MB-231 cells treated with indicated drugs. (B) RT-qPCR analysis of *FYN* expression levels in MDA-MB-231 cells treated with indicated drugs (n=3). (C) RT-qPCR analysis of indicated jumonji family histone demethylase expression levels (n=3). (D) Changes in *FYN* mRNA levels upon NVP-ADW742 treatment in MDA-MB-231 Cas9 cells expressing indicated sgRNAs (n=4). (E) *KDM4A* mRNA levels in primary tumor tissues of indicated subtypes in GSE25066 cohort. (F) Positive correlation of *FYN* and *KDM4A* mRNA levels in CCLE database. (G) western blot analysis of MDA-MB-231 Cas9 cells expressing indicated sgRNA and treated with indicated drugs. (H) MTT assay of MDA-MB-231 Cas9 cells expressing indicated sgRNAs and treated with indicated drugs (n=3). (I) SynergyFinder analysis of MDA-MB-231 cells treated with indicated drug combinations (n=2). (J-K) western blot analysis (J) and RT-qPCR analysis (K) of MDA-MB-231 cells treated with indicated drugs (n=3). (L) mRNA expression levels of SRC family kinases in breast cancer stem cells treated with QC6352 in RNA sequencing data described in Metzger et. al. (29) (M) H3K9me3 and *KDM4A* enrichment at genomic locus encoding *FYN* promoter in ChIP sequencing data described in the same study as (L). (N) H3K9me3 Chromatin immunoprecipitation-qPCR analysis of MDA-MB-231 cells treated with indicated drug at specified genomic loci (n=3). All data are plotted as mean±s.d. Unpaired two-sided Student's t-test in B,C,D,E,H and K. Paired two-sided Student's t-test in N. *, p<0.05; **, p<0.01; ***, p<0.001; n.s., p>0.05. All replicates are biological replicates.

MDA-MB-231 xenograft tumor growth without causing overt changes in animal health (Figs. 4B and S6B). Additionally, the expression levels of *FYN* and *KDM4A* were found to be correlated with poor prognosis in a previously reported breast cancer cohort(19), highlighting the potential of targeting these two genes as therapeutic targets for TNBC (Fig. 4C). Collectively, our results demonstrate that upregulation of *KDM4* upon TKI treatment reduces H3K9me3 mark in *FYN* enhancer, thereby increasing *FYN* expression and promoting cell survival under TKI treatment (Fig. 4D).

The observed epigenetic alterations in regulators conferring resistance to multiple cancer drugs closely resemble non-genetic changes associated with the generation of drug-tolerant persisters(26). Indeed, prolonged incubation of MDA-MB-231 cells treated with TKIs or conventional chemotherapy drugs such as doxorubicin or paclitaxel resulted in increased levels of *FYN* (Fig. 5A). Curiously, *KDM4A* expression was only upregulated upon treatment with NVP-ADW742 and gefitinib, suggesting that while *FYN* upregulation is a general feature of drug tolerant cells, the mechanism of *FYN* upregulation may vary depending on the specific drug being used. Analysis of previously published RNA sequencing data from a series of Osimertinib tolerant EGFR mutated lung cancer cell lines(31) revealed higher expression levels of *FYN* and *KDM4A* in the drug persisters, but not SRC (Fig. 5B). Consistently, we confirmed upregulation of *FYN* at both the protein and mRNA levels in gefitinib and osimertinib resistant PC9 and HCC827 cells (Figs. 5C-D). Pharmacological inhibition of *FYN* or downregulation of *FYN* expression through inhibition of *KDM4* sensitized gefitinib resistant PC9 cells to EGFR inhibitor, suggesting that *FYN-KDM4* are responsible for gefitinib resistant phenotype in this cell line (Figs. 5E-G).

Importantly, upregulation of *FYN* has been consistently observed in multiple independent studies involving drug-tolerant cancer cell lines and patient-derived xenografts treated with various drugs that have distinct target profiles, including TKIs (e.g., lapatinib, a HER2 inhibitor) and chemotherapy drugs (e.g., irinotecan) (Fig. 5H). Moreover, enrichment of *FYN* has also been observed in residual disease following chemotherapy, indicating its potential role in mediating drug tolerance during chemotherapy (Fig. 5I). Notably, an analogous increase in *KDM4* was not consistently observed across all tumor models tested in Figures 5H-I (Fig. S7A-B). This suggests that, as previously noted in Figure 5A, while *FYN* serves as a general mediator of drug tolerance, the specific mechanisms underlying its upregulation may vary depending on the cancer type and the drug being administered. Taken together, these lines of evidence further support our findings in TNBC cell lines and suggest that *FYN* acts as a common mediator of drug tolerance.

Discussion

In this study, we employed massively parallel, combinatorial CRISPR screening to identify combinations of TKIs that exhibit synergistic effects in eliminating triple-negative breast cancer (TNBC). We discovered and validated that concurrent targeting of *FYN*, along with other tyrosine kinases such as IGF1R, EGFR or ABL2 can synergistically eradicate TNBC and impede cancer growth. Our findings also provide evidence that the transcriptional upregulation of *FYN*, facilitated by the activation of *KDM4* histone demethylases, confers resistance and persistence to TKIs. Upregulation of *FYN* is a general feature of drug tolerant cancer cells, suggesting *FYN* as an attractive target for minimization of tumor recurrence after treatment.

This research provides basis for breakthrough combinatorial therapy achieving effective targeted therapy with minimal risk of developing resistance. Our combinatorial CRISPR screening demonstrates that treatment with TKIs or histone demethylase inhibitors enhances the sensitivity of cells to other TKIs. Consequently, drug combinations exhibit a more potent inhibition of cancer growth than the simple sum of the therapeutic effects of individual drugs. Furthermore, synergistic drug combinations enable a reduction in the dosage of each drug, with minimal compromise in therapeutic efficacy. Such combinations yield a therapeutic response comparable

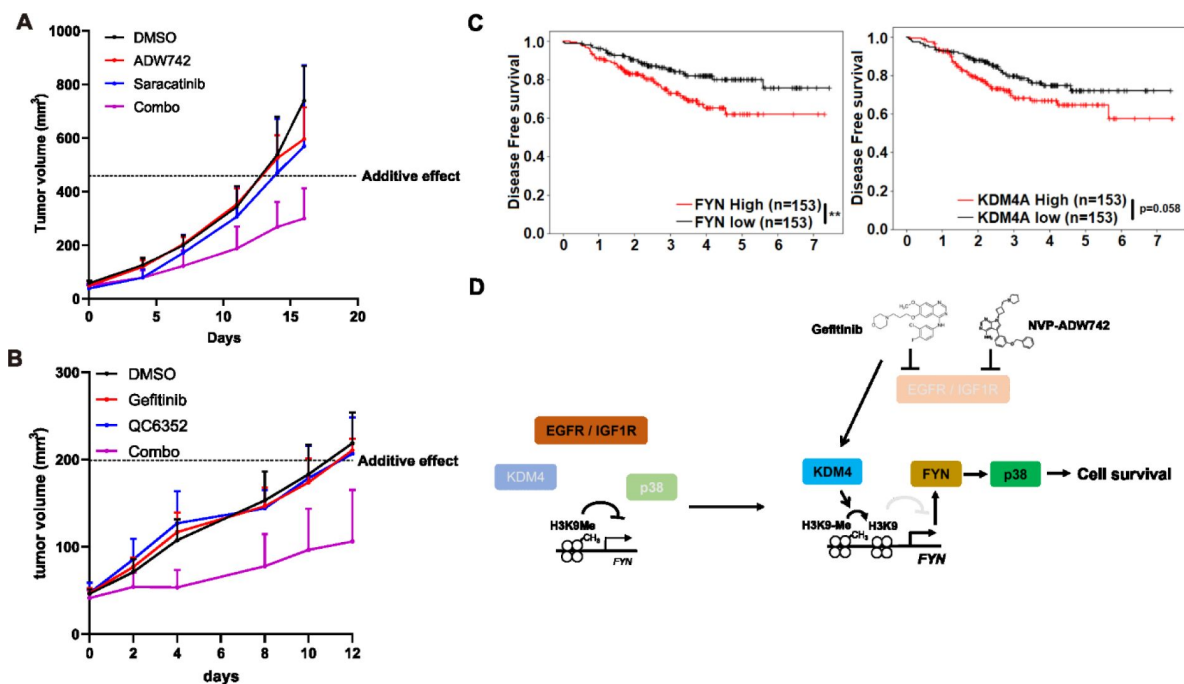


Figure 4.

Combination therapy targeting FYN+IGF1R and KDM4+EGFR synergistically eliminates tumor *in vivo*.

(A-B) Tumor volume for MDA-MB-231 xenografts treated with indicated drugs. Additive effects were calculated by Bliss independence model (n=5). (C) Distant relapse free survival of GSE25066 patient cohort classified by *FYN* (left) and *KDM4A* (right) mRNA expression. (D) Schematic diagram of the mechanism of KDM4-FYN conferring TKI resistance. All data are plotted as mean±s.d. *, p<0.05; **, p<0.01; ***, p<0.001; n.s., p>0.05. All replicates are biological replicates.

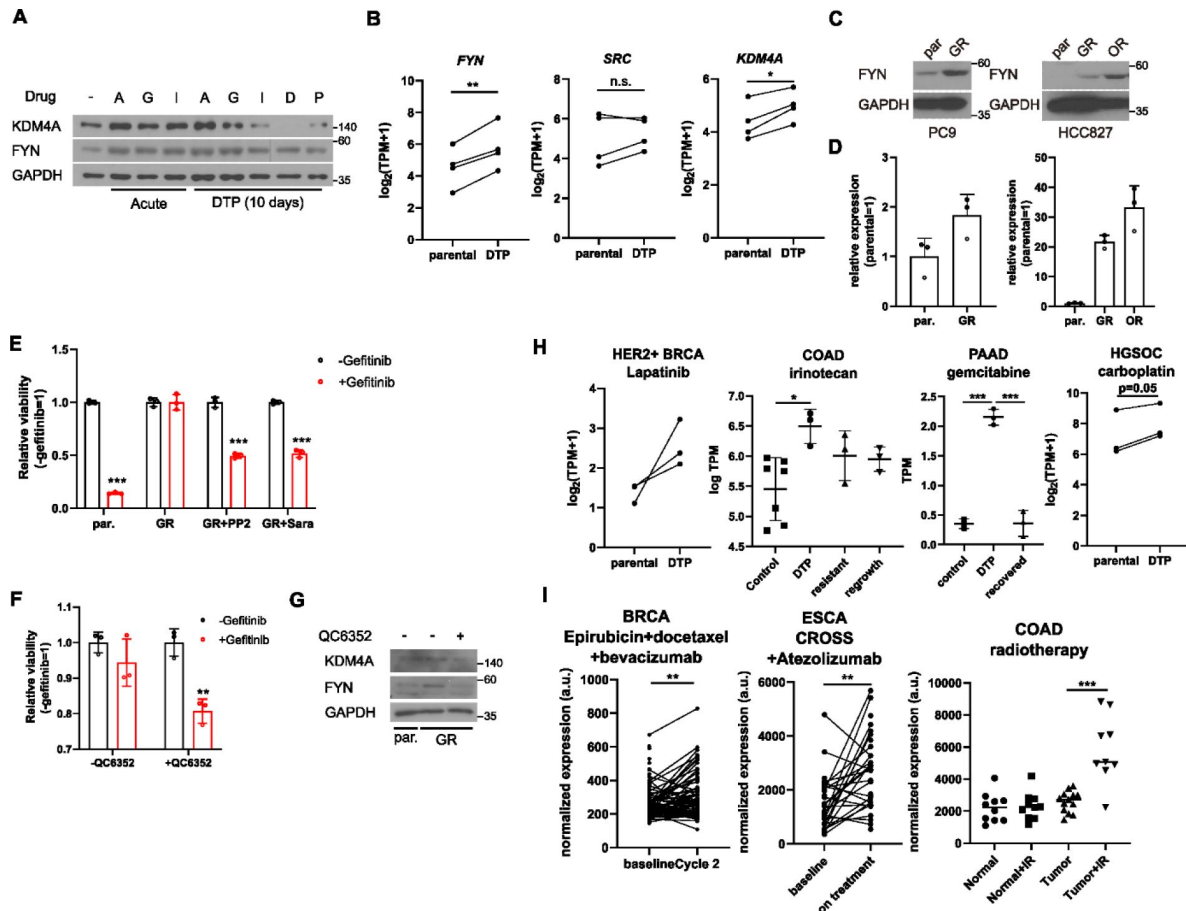


Figure 5.

***FYN* and *KDM4* are associated with drug tolerance.**

(A) MDA-MB-231 cells treated with indicated drugs for short (acute: 2 days) and long (DTP: 10 days) time periods. (B) Summary of mRNA expressions of indicated genes in EGFR mutant lung cancer cells (parental) and their derivative Osimertinib tolerant persisters (DTP). (C-D) western blots (C) and RT-qPCR (D) analyses of indicated parental and EGFR inhibitor resistant lung cancer cells. (E) *FYN* mRNA expression levels of parental and DTP populations in various cancers treated with indicated drugs (n=3). (F) *FYN* mRNA expression levels of residual disease after indicated treatments (n=3). (G-H) MTT assay with PC9 parental (par.) and gefitinib resistant (GR) cells treated with indicated drug combinations. (I) western blot analysis with PC9 cells treated with QC6352. All data are plotted as mean±s.d. Paired two-sided Student's t-test in B, E (HER2+BRCA set and HGSOC carboplatin set), and F (BRCA set and ESCA set), and unpaired two-sided Student's t-test in E (COAD and PAAD sets) and F (COAD set). *, p<0.05; **, p<0.01; ***, p<0.001; n.s., p>0.05. All replicates are biological replicates.

to that achieved with significantly higher doses of each individual agent. We anticipate that combinatorial therapy has the potential to mitigate side effects by minimizing the dosage of each drug, thus widening the therapeutic window.

Additionally, our work underscores the potential utility of kinase inhibitors with promiscuous binding profiles. Most kinase inhibitors target the highly conserved ATP binding pocket, resulting in multiple target interactions(32 [↗](#)). While substantial efforts have been made to enhance the specificity of kinase inhibitors, the repurposing of inhibitors with more relaxed specificity as dual or multi-kinase inhibitors, targeting synergistic kinase targets, may offer therapeutic advantages. The rational design of dual kinase inhibitors holds significant promise for advancing therapeutic interventions.

It is intriguing to observe that *FYN* is specifically upregulated in various models of drug resistance and tolerance. While other SRC family kinases have been linked to drug resistance(33 [↗](#)), the precise molecular mechanisms underlying their increased contribution to cell survival upon drug treatment remained unclear. Our findings reveal that *FYN* is specifically upregulated at the mRNA level through epigenetic regulations, providing further depth to our understanding of drug resistance in cancer therapy.

Furthermore, our work highlights the significance of histone demethylases in TKI resistance. Numerous histone demethylases have been implicated in drug resistance and tolerance across different cancer drug types. For instance, the KDM5 family of H3K4 demethylases has been associated with the drug-tolerant persister phenotype against multiple TKIs(26 [↗](#)). In our study, we identify *KDM4* as a critical factor in the generation of drug-tolerant persisters in breast cancer. *KDM4* is known to be upregulated in various cancers, including breast cancer, and promotes key malignant traits. Previous studies have demonstrated the essential role of KDM4 in induced pluripotency through its interaction with pluripotency factors(34 [↗](#)). These findings suggest that an KDM4 inhibitor could be a promising therapeutic target with specific activity against cancer stem cells. Consistently, specific inhibitors targeting KDM4 have recently been developed and shown to inhibit the generation of breast cancer stem cells(29 [↗](#)). Given our discoveries regarding the involvement of *KDM4* in drug resistance in breast and lung cancer, the development of novel drugs targeting KDM4 holds significant therapeutic potential.

Materials and methods

Cell Culture

HEK293T, MDA-MB-231, Hs578T cells were obtained from American Type Cell Culture (ATCC) HCC1143, HCC1395, HCC1937 were obtained from Korean cell line bank. HEK293T and MDA-MB-231 were grown in DMEM (Gibco) supplemented with 10% FBS (Corning) and penicillin/streptomycin (Gibco). Hs578T and HCC1143, HCC1395, HCC1937 were grown in RPMI1640 (Gibco) supplemented with 10% FBS.

Combinatorial library construction

Combinatorial library was constructed as previously described(15 [↗](#)). The sgRNAs used in the screens were cloned in pAWp28 storage vector in two versions: one version containing human U6 driven sgRNA with wild type scaffold, and another containing mouse U6 driven sgRNA with cr2 variant scaffold. The sgRNA expression cassette consisting of U6 promoter and sgRNA were subject to one-pot, iterative cloning into lentiviral pTK799 vector using BglII-MfeI restriction sites flanking the sgRNA expression cassette and BamHI-EcoRI sites in pTK799. pTK799 vector is derived from pAWp12(15 [↗](#)) by replacing CMV-GFP selectable marker to EFS-Puro.

Combinatorial CRISPR screening procedure

Lentivirus was generated in HEK293T cells by transfecting lentiviral transfer vector, and helper vectors (psPAX2, and pVSV-G) using Fugene HD (Promega). Lentiviral supernatant was collected 48 hours after transfection, and was frozen and stored in -80C. The appropriate titer for lentiviral infection was determined by infecting MDA-MB231 cells with two-fold serial dilution of lentiviral supernatant, selecting with puromycin 2 days after infection for 2 days, and determining cell viability with AQuaeous one cell viability, MTS assay (Promega). After determining the titer of lentiviral supernatant, 100 million MDA-MB231 cells carrying constitutively expressed Cas9 were infected with CombiGEM library at MOI of 0.3. The expected initial coverage is $100 \text{ million} \times 0.3 / (54,289 \text{ different sgRNA combinations}) = 553$. Three days after infection, the infected cells were either harvested as day 0 sample or selected with 2 μ g/mL puromycin. Cells were treated with benzonase before harvesting to minimize carryover of plasmid DNA in lentiviral supernatant. Cells were grown in the presence of 2 μ g/mL puromycin for 20 days before harvesting.

The genomic DNA of harvested cells were isolated using Blood & Cell Culture Maxi kit (Qiagen). The PCR amplicon spanning the two sgRNAs were generated with PCR using Q5 High Fidelity DNA polymerase (New England Biolabs) and the following primers:

F: CAAGCAGAAGACGGCATACGAGAT CCTAGTAAGTATAGAGGCTTAATGTGCG

R: AATGATACGGCGACCACCGAGATCTACAC NNNNNN ACACGAATTCTGCGTGGATCCAA

The six variable nucleotides were added in reverse primer for multiplexing.

The PCR protocols involves 60 seconds of initial DNA denaturation at 98C, and 20 cycles of 10 seconds denaturation at 98C, 10 seconds annealing at 67C, and 120 seconds elongation at 72C. All genomic DNA isolated were used in PCR reaction at concentration of 40 μ g/mL. All PCR products were combined and precipitated with isopropanol at room temperature. The precipitated DNA was resuspended in 400 μ L EB buffer (Qiagen) and gel purified. The purified PCR products were sent for NGS by NextSeq500 paired end sequencing with the following sequencing primers:

Forward read: GGACTAGCCTTATTTGAACTTGCTATGCAGCTTTCTGCTTAGCTCTCAAAAC

Forward index read:

CGGTGCCACTTTTCAAGTTGATAACGGACTAGCCTTATTTAACTTGCTATTTCTAGCTCTAAAAAC

Reverse read: GCA CCG AGT CGG TGC TTT TTT GGA TCC ACG GCA GAA TTC GTGT

Data analysis

The sgRNA sequences were identified and their occurrences were counted with C++ script deposited in Github (<https://github.com/tackhoonkim/combinatorial-CRISPR-screens-2023>).

Validation of screens using sgRNAs

Individual sgRNA was cloned to either pTK1329, and pTK1336 that are both derived from pAWp12 with EFS-GFP and EFS-mCherry, respectively, as selectable markers. Validation of synthetic lethality between gene A and B were analyzed by infecting MDA-MB-231 Cas9 cells with four combinations of lentiviral supernatant pairs (MOI ~ 0.5 each) containing (1) GFP-sgA and mCherry-sgB; (2) GFP-sgA and mCherry-sgCon; (3) GFP-sgCon and mCherry-sgB; (4) GFP-sgCon and mCherry-sgCon. The fraction of GFP/mCherry double positive cells were analyzed using

BD Accuri C6 and its accompanying software. The expected fold change in sgA+sgB were calculated as $FC_{sgA+sgCon} \times FC_{sgCon+sgB}$, where FC is normalized fold change in fraction of GFP/mCherry double positive cells relative to those infected with GFP-sgCon and mCherry-sgCon.

MTT Cell viability assay

Cells were seeded at 1000-2000 cells/well in 96 well plate. Tyrosine kinase inhibitors at indicated combination of dose were treated 12 hours after seeding, and cells were grown for 3 days. The relative viability was measured by EzCyttox cell viability assay (Dojindo). The absorbance at 450nm wavelength was measured using EnVision multimode plate reader (PerkinElmer).

Cell death and cell proliferation assay

Cells were incubated with tyrosine kinase inhibitors for 48 hours. Cell proliferation was quantified with BrdU assay using FITC conjugated BrdU antibody (Biolegend) and propidium iodide/RNase A solution (Cell Signaling), analyzed with BD Accuri C6 and accompanied software. Cell death was quantified with Live-Dead cell staining kit (Molecular Probes) by flow cytometry analysis using BD Accuri C6 and accompanied software.

Western blot analysis

Cells were treated with drugs for 48 hours unless otherwise indicated. Cells were lysed in RIPA buffer supplemented with protease inhibitor and phosphatase inhibitor cocktail. Antibodies used for western blot analysis were all from Cell Signaling Technology. The original blot scans are available in figure S8.

Xenograft assay

All animal experiments were approved by IACUC of Korea Institute of Science and Technology (KIST). Six week old female nude mice were injected with 5×10^6 MDA-MB-231 cells suspended in 1:1 (w/w) mixture of PBS and growth factor reduced Matrigel (Corning) in fourth inguinal mammary fat pad. Starting two weeks after tumor cell injection, saracatinib (50mg/kg mouse body weight), NVP-ADW742 (20mg/kg), gefitinib (20mg/kg), QC6352 (10mg/kg) in 45% saline+40% polyethyleneglycol 300 (sigma)+5% Tween-80 (sigma)+5% DMSO (sigma) were injected intraperitoneally every 24 hours for two weeks. Tumor volume was measured by digital caliper and calculated as $(width)^2 \times length \times 0.5$.

Public database analyses

Gene Expression Omnibus (GEO) data with breast cancer cohort (GSE25066(19)) were analyzed using web based platform Cancer Target Gene Screening (<https://ctgs.biohackers.net>)(35). Cancer Cell Line Encyclopedia (CCLE) data were analyzed using depmap R package version 1.14. The list of GEO data used for analysis are listed in table S2.

Primary TNBC organoid culture and drug treatment

Tumor tissue were collected in breast cancer at the National Cancer Center (Goyang, Republic of Korea) with IRB approval. After additional refining steps and cell counting, 1×10^5 cells were embedded in 50 μ L of Matrigel (Corning, NY, USA) and seeded in each well of a 24-well cell culture plate. After the matrigel was solidified, 500 μ L medium supplemented with defined growth factors as described by Clevers and colleagues(36), was added to each well and grown under standard culture conditions (37 °C, 5% CO₂).

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Author contributions

T.K., B.-S.P., H.J. and J.K. performed experiments. T.K. and T.K.L. supervised the research. S.H. S.-Y.J. and S.Y.K. performed experiments with TNBC patient derived organoids. D.K., S.K.L. generated and provided Osimertinib resistant HCC827 cell line.

Ethics Statement

All experiments with human tumor organoids were conducted in accordance with the requirements of the National Cancer Center Institutional Review Board (IRB).

Availability of data and materials

The NGS data for CRISPR screening results are available under NCBI SRA accession code PRJNA976939.

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Reviewer #1 (Public Review):

Summary:

The authors employed a combinatorial CRISPR-Cas9 knockout screen to uncover synthetically lethal kinase genes that could play a role in drug resistance to kinase inhibitors in triple-negative breast cancer. The study successfully reveals FYN as a mediator of resistance to depletion and inhibition of various tyrosine kinases, notably EGFR, IGF-1R, and ABL, in triple-negative breast cancer cells and xenografts. Mechanistically, they demonstrate that KDM4 contributes to the upregulation of FYN and thereby is an important mediator of drug resistance. All together, these findings suggest FYN and KDM4A as potential targets for combination therapy with kinase inhibitors in triple-negative breast cancer. Moreover, the study may also have important implications for other cancer types and other inhibitors, as the authors suggest that FYN could be a general feature of drug-tolerant persister cells.

Strengths:

(1) The authors used a large combination matrix of druggable tyrosine kinase gene knockouts, enabling studying of co-dependence of kinase genes. This approach mitigates off-target effects typically associated with kinase inhibitors, enhancing the precision of the findings.

(2) The authors demonstrate the importance of FYN in drug resistance in multiple ways. They demonstrate synergistic interactions using both knockouts and inhibitors, while also revealing its transcriptional upregulation upon treatment, strengthening the conclusion that FYN plays a role in the resistance.

(3) The study extends its impact by demonstrating the potent *in vivo* efficacy of certain combination treatments, underscoring the clinical relevance of the identified strategies.

Weaknesses:

(1) The methods and figure legends are incomplete, posing a barrier to the reproducibility of the study and hindering a comprehensive understanding and accurate interpretation of the results.

(2) The authors make use of a large quantity of public data (Fig. 2D/E, Fig. 3F/L/M, Fig 4C, Fig 5B/H/I), whereas it would have strengthened the paper to perform these experiments themselves. While some of this data would be hard to generate (e.g. patient data) other data could have been generated by the authors. The disadvantage of the use of public data is that it merely comprises associations, but does not have causal/functional results (e.g. FYN inhibition in the different cancer models with various drugs). Moreover, by cherry-picking the data from public sources, the context of these sources is not clear to the reader, and thus harder to interpret correctly. For example, it is not directly clear whether the upregulation of FYN in these models is a very selective event or whether it is part of a very large epigenetic re-programming, where other genes may be more critical. While some of the used data are from well-known curated databases, others are from individual papers that the reader should assess critically in order to interpret the data. Sometimes the public data was redundant, as the authors did do the experiments themselves (e.g. lung cancer drug-tolerant persisters), in this case, the public data could also be left out.

More importantly, the original sources are not properly cited. While the GEO accession numbers are shown in a supplementary table, the articles corresponding to this data should be cited in the main text, and preferably also in the figure legend, to clarify that this data is from public sources, which is now not always the case (e.g. line 224-226). If these original papers do already mention the upregulation of FYN, and the findings from the authors are thus not original, these findings should be discussed in the Discussion section instead of shown in the Results.

(3) The claim in the abstract (and discussion) that the study "highlights FYN as broadly applicable mediator of therapy resistance and persistence", is not sufficiently supported by the results. The current study only shows functional evidence for this for an EGFR, IGF1R, and Abl inhibitor in TNBC cells. Further, it demonstrates (to a limited extent) the role of FYN in gefitinib and osimertinib resistance (also EGFR inhibitors) in lung cancer cells. Thus, the causal evidence provided is only limited to a select subset of tyrosine kinase inhibitors in two cancer types. While the authors show associations between FYN and drug resistance in other cancer types and after other treatments, these associations are not solid evidence for a causal connection as mentioned in this statement. Epigenetic reprogramming causing drug resistance can be accompanied by altered gene expression of many genes, and the upregulation of FYN may be a consequence, but not a cause of the drug resistance. Therefore, the authors should be more cautious in making such statements about the broad applicability of FYN as a mediator of therapy resistance.

(4) The rationale for picking and validating FYN as the main candidate gene over other genes such as FGFR2, FRK2, and TEK is not clear.

a. While gene pairs containing FGFR2 knockouts seemed to be equally effective as FYN gene pairs in the primary screening, these could not be validated in the validation experiment. It is unclear whether multiple individual or a pool of gRNAs were used for this validation, or whether only 1 gRNA sequence was picked per gene for this validation. If only 1 gRNA per gene was used, this likely would have resulted in variable knockout efficiencies. Moreover, the T7 endonuclease assay may not have been the best method to check knockout efficiency, as it only implies endonuclease activity around a gene (but not to the extent of indels that can cause frameshifts, such as by TIDE analysis, or extent of reduction in protein levels by western blot).

b. Moreover, FRK2 and TEK, also demonstrated many synergistic gene pairs in the primary screen. However, many of these gene pairs were not included in the validation screening. The selection criteria of candidate gene pairs for validation screening is not clear. Still, TEK-ABL2 was also validated as a strong hit in the validation screen. The authors should better explain the choice of FYN over other hits, and/or mention that TEK and FRK2 may also be important targets for combination treatment that can be further elucidated.

(5) On several occasions, the right controls (individual treatments, performed in parallel) are not included in the figures. The authors should include the responses to each of the single treatments, and/or better explain the normalization that might explain why the controls are not shown.

a. Figure 2G: The effect of PP2 treatment, without combined treatment, is not shown.

b. Figure 2H/3G: The effect of the knockouts on growth alone, compared to sgGFP, is not demonstrated. It is unclear whether the viability of knockouts is normalized to sgGFP, or to each untreated knockout.

c. Figure 2L: The effect of SB203580 as a single treatment is not shown.

(6) The study examines the effects at a single, relatively late time point after treatment with inhibitors, without confirming the sequential impact on KDM4A and FYN. The proposed sequence of transcriptional upregulation of KDM4A followed by epigenetic modifications leading to FYN upregulation would be more compellingly supported by demonstrating a consecutive, rather than simultaneous, occurrence of these events. Furthermore, the protein level assessment at 48 hours (for RNA levels not clearly described), raises concerns about potential confounding factors. At this late time point, reduced cell viability due to the combination treatment could contribute to observed effects such as altered FYN expression and P38 MAPK phosphorylation, making it challenging to attribute these changes solely to the specific and selective reduction of FYN expression by KDM4A.

(7) The cut-off for considering interactions "synergistic" is quite low. The manual of the used "SynergyFinder" tool itself recommends values above >10 as synergistic and between -10 and 10 as additive (https://synergyfinder.fimm.fi/synergy/synfin_docs/). Here, values between 5-10 are also considered synergistic. Caution should be taken when discussing those results. Showing the actual dose response (including responses to each single treatment) may be required to enable the reader to critically assess the synergy, along with its standard deviation.

(8) As the effect size on Western blots is quite limited and sometimes accompanied by differences in loading control, these data should be further supported by quantifications of signal intensities of at least 3 biological replicates (e.g. especially Figure 3A/5A). The figure legends should also state how many independent experiments the blots are representative of.

(9) While the article provides mechanistic insights into the likely upregulation of FYN by KDM4A, this constitutes only a fragment of the broader mechanism underlying drug resistance associated with FYN. The study falls short in investigating the causes of KDM4A

upregulation and fails to explore the downstream effects (except for p38 MAPK phosphorylation, which may not be complete) of FYN upregulation that could potentially drive sustained cell proliferation and survival. These omissions limit the comprehensive understanding of the complete molecular pathway, and the discussion section does not address potential implications or pathways beyond the identified KDM4A-FYN axis. A more thorough exploration of these aspects would enhance the study's contribution to the field.

(10) FYN has been implied in drug resistance previously, and other mechanisms of its upregulation, as well as downstream consequences, have been described previously. These were not evaluated in this paper, and are also not discussed in the discussion section. Moreover, the authors did not investigate whether any of the many other mechanisms of drug resistance to EGFR, IGF1R, and Abl inhibitors that have been described, could be related to FYN as well. A more comprehensive examination of existing literature and consideration of alternative or parallel mechanisms in the discussion would enhance the paper's contribution to understanding FYN's involvement in drug resistance.

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Reviewer #2 (Public Review):

Summary:

Kim et al. conducted a study in which they selected 76 tyrosine kinases and performed CRISPR/Cas9 combinatorial screening to target 3003 genes in Triple-negative breast cancer (TNBC) cells. Their investigation revealed a significant correlation between the FYN gene and the proliferation and death of breast cancer cells. The authors demonstrated that depleting FYN and using FYN inhibitors, in combination with TKIs, synergistically suppressed the growth of breast cancer tumor cells. They observed that TKIs upregulate the levels of FYN and the histone demethylase family, particularly KDM4, promoting FYN expression. The authors further showed that KDM4 weakens the H3K9me3 mark in the FYN enhancer region, and the inhibitor QC6352 effectively inhibits this process, leading to a synergistic induction of apoptosis in breast cancer cells along with TKIs. Additionally, the authors discovered that FYN is upregulated in various drug-resistant cancer cells, and inhibitors targeting FYN, such as PP2, sensitize drug-resistant cells to EGFR inhibitors.

Strengths:

This study provides new insights into the roles and mechanisms of FYN and KDM4 in tumor cell resistance.

Weaknesses:

It is important to note that previous studies have also implicated FYN as a potential key factor in drug resistance of tumor cells, including breast cancer cells. While the current study is comprehensive and provides a rich dataset, certain experiments could be refined, and the logical structure could be more rigorous. For instance, the rationale behind selecting FYN, KDM4, and KDM4A as the focus of the study could be more thoroughly justified.

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